

Zero-Shot Cross-Lingual Audio Deepfake Detection on Brazilian Portuguese Speech

João Victor
Universidade Federal de Goiás (UFG) / CEIA
Goiânia, Brazil
TODO@ufg.br

Abstract

TODO: abstract.

Keywords

audio deepfake detection, anti-spoofing, cross-lingual, wav2vec 2.0, XLS-R, explainability, Portuguese

1 Introduction

TODO.

2 Related Work

2.1 Self-Supervised Front-Ends for Anti-Spoofing

Tak et al. [4] TODO.

2.2 Robustness and Real-World Generalization

Channing et al. [1] TODO.

2.3 Explainability in Audio Deepfake Detection

Grinberg et al. [3] and Fullwood and Monroe [2] TODO.

3 Methodology

3.1 Pre-processing

TODO.

3.2 Model

TODO.

4 Experimental Setup

TODO: descrever o BRSpeech-DF (tamanho, origem, como o rótulo bonafide/spoof é resolvido), hardware usado, e re-confirmar que a versão final rodou no ambiente único definido (ver pendência nas instruções do projeto).

5 Results

5.1 Zero-Shot Performance

TODO.

5.2 Robustness Under Real-World Degradations

TODO.

Table 1: Experimental configuration.

Parameter	Value
Sample rate	16,000 Hz
Fixed window	64,600 samples (≈ 4.04 s)
Checkpoint	nii-yamagishilab/mms-300m-anti-deepfake
Dataset	AKCIT-Deepfake/BRSpeech-DF (test split)
Samples per class	200 (balanced)
Seed	42

6 Explainability

6.1 GATR: Time-Domain Relevancy

TODO.

6.2 Occlusion Analysis: Speech vs. Silence

TODO.

7 Discussion and Limitations

TODO.

8 Conclusion and Future Work

TODO.

References

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